

Navigating Exact-Response Fibres in Shor Order Finding

Reconstructed manuscript source, revision v0.9.4-m4

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Reconstructed source prepared from archived v0.9.4-m3 PDF and revised against v0.9.4-r1
corrective results

Source provenance. The editable source of manuscript revision v0.9.4-m3 was unavailable. The v0.9.4-m4 LaTeX source was reconstructed from the archived m3 PDF and then revised against the frozen v0.9.4-r1 corrective result. It must not be interpreted as the recovered original m3 source.

Abstract

We study a synthetic, exact state-vector model of Shor order finding in which the ideal phase-estimation response is held fixed while implementation-noise coordinates are varied along a four-dimensional exact-response fibre. On a frozen six-instance benchmark with 72 meta-records, intrinsic Euclidean ascent on this fibre improves held-out exact-order recovery relative to the unchanged reference, a basis-invariant random tangent comparator, and an SPSA comparator run under the same composite response-call ledger. In the v0.9.4-r1 corrective archive, the equal-instance mean gains are $4.937962211018602 \times 10^{-4}$ over reference, $3.411545293638851 \times 10^{-4}$ over random search, and $1.2279600467454457 \times 10^{-4}$ over SPSA, with paired hierarchical bootstrap 95% confidence intervals. In comparator order (reference, random, SPSA), these are

$$\begin{aligned} &[2.902978897001541 \times 10^{-4}, 6.822119290361299 \times 10^{-4}], \\ &[2.3521176375243113 \times 10^{-4}, 4.5502493973362636 \times 10^{-4}], \\ &[5.996365059637254 \times 10^{-5}, 2.0092595128065157 \times 10^{-4}]. \end{aligned}$$

The intrinsic method beats SPSA on 72/72 records and has positive instance-level means on 6/6 instances; it beats the random comparator on 68/72 records and has positive instance-level means on 6/6 instances. These results support navigability of the stipulated fibre under the declared synthetic noise family only. They are not evidence of a native hardware advantage, fault-tolerant resource reduction, cryptographic-scale factoring, practical computational advantage, or an asymptotic improvement to Shor’s algorithm.

1 Introduction

Shor order finding reduces factoring to estimating the period of modular exponentiation. In the ideal algorithm, the relevant response is the phase-estimation distribution induced by an order r and a control register of size $M = 2^m$. Practical implementations, however, introduce coherent and stochastic distortions before the exact-order decision rule is applied. The question addressed here is narrow: within a declared non-native synthetic noise ansatz, can one move in directions that preserve the ideal response while improving the held-out exact-order recovery probability?

The experiment studies exact-response fibres rather than new asymptotic algorithms. A fibre is the set of parameter perturbations that leave the declared ideal response invariant to first order.

The intrinsic method moves inside a four-dimensional tangent fibre and optimizes the held-out task objective. The frozen benchmark compares it with an unchanged reference, basis-invariant random tangent search, and SPSA. Revision v0.9.4-r1 is a reproducibility correction: it pins the analytic tangent-coordinate basis used by SPSA. The original v0.9.4 tangent subspace was correct, but the concrete coordinate basis used inside the degenerate four-dimensional subspace was not archived; this matters for coordinate Rademacher perturbations.

2 Model And Exact-Response Fibre

2.1 Phase-Estimation Response

For an instance (N, a) with multiplicative order r , the ideal phase-estimation amplitudes are indexed by a control-register outcome $y \in \{0, \dots, M-1\}$ and by an eigensector s . The reconstructed formula uses the normalization implemented by the archive:

$$A_s(y; \theta, \xi) = \frac{1}{M} \sum_{x=0}^{M-1} \exp \left[2\pi i x \left(\frac{s}{r} - \frac{y}{M} \right) + i\phi_x(\theta, \xi) \right]. \quad (1)$$

The exact-order success probability is obtained by summing the probabilities of outcomes whose continued-fraction post-processing recovers the exact order. The code retains final normalization as numerical insurance, not as a mathematical correction to the amplitude definition.

2.2 Synthetic Implementation Ansatz

The archive uses a shared nine-parameter synthetic implementation/noise ansatz. Its declared response matrix has rank five and a four-dimensional kernel. The sixth response coordinate is an explicit consistency coordinate,

$$r_6 = r_1 + r_2 + r_3 + r_5. \quad (2)$$

Consequently, the tangent subspace has dimension four. This subspace is the object preserved by the original v0.9.4 archive.

2.3 Pinned Analytic Tangent Basis

The corrective r1 archive makes the SPSA coordinate convention explicit. The tangent basis is fixed to the following orthonormal columns in $\mathbb{R}^9$:

$$t_1 = (e_0 + e_3)/\sqrt{2}, \quad (3)$$

$$t_2 = (e_1 - e_5)/\sqrt{2}, \quad (4)$$

$$t_3 = (e_1 + e_5 - 2e_6)/\sqrt{6}, \quad (5)$$

$$t_4 = (e_2 - e_8)/\sqrt{2}. \quad (6)$$

The deposited basis hash is 03236c148c7dc761344d9bed57d907e1494cd9ce0b4e7c9e6d043d12d257b224. The archived residuals are $2.4655887277772096 \times 10^{-16}$ for the Gram check, 0 for the response check, and $3.895959316580178 \times 10^{-16}$ for the projector check.

3 Protocol

The frozen instance family is (15, 2), (15, 7), (21, 2), (21, 5), (33, 2), and (35, 2), with exact orders 4, 4, 6, 6, 10, and 12. Each instance has 12 meta-seeds. Each record uses five training seeds and eight shifted held-out seeds. The primary statistic is the equal-instance-weight mean held-out gain, with paired hierarchical bootstrap intervals obtained by resampling instances and then meta-records. All methods are compared under an equal composite response-call ledger of 5520 calls per record. The intrinsic method uses 5500 task calls plus 20 padding calls, whereas the random and SPSA baselines use 5520 task calls.

Table 1: Composite ledger per record.

Method	Task calls	Response/padding calls	Composite total
Intrinsic	5500	20	5520
Random	5520	0	5520
SPSA	5520	0	5520

4 Primary Results

Table 2 reports the r1 aggregate outcomes. All 13 primary gates in the corrective archive pass. The maximum three-way budget gap is 0. The remaining maximum audit residuals are

$$\begin{aligned} \text{intrinsic response} &= 5.2958809290937725 \times 10^{-17}, \\ \text{ideal-multiplier change} &= 7.850462293418876 \times 10^{-17}, \\ \text{random normal component} &= 1.265777190428121 \times 10^{-15}. \end{aligned}$$

Table 2: Primary aggregate held-out gains in the v0.9.4-r1 corrective archive.

Comparison	Mean gain	95% CI low	95% CI high	Record wins
Intrinsic vs reference	0.0004937962211018602	0.0002902978897001541	0.0006822119290361299	72/72
Intrinsic vs random	0.0003411545293638851	0.00023521176375243113	0.00045502493973362636	68/72
Intrinsic vs SPSA	0.00012279600467454457	0.00005996365059637254	0.00020092595128065157	72/72

Table 3: Frozen instances and exact-order success bins.

Instance	Order	Control qubits	Success bins	Ideal probability
$N15, a2$	4	4	4, 12	0.5000000000000000
$N15, a7$	4	4	4, 12	0.5000000000000000
$N21, a2$	6	7	21, 107	0.228072894737454
$N21, a5$	6	7	21, 107	0.228072894737454
$N33, a2$	10	7	13, 115	0.175232715792459
$N35, a2$	12	8	21, 107, 149, 235	0.228072894737451

Table 4: Per-instance mean gains and win counts in r1.

Instance	vs reference	vs random	vs SPSA	Random wins	SPSA wins
$N15, a2$	0.000695540898	0.000357417952	0.000201403988	10/12	12/12
$N15, a7$	0.000764201566	0.000291541463	0.000070793081	12/12	12/12
$N21, a2$	0.000194208274	0.000230987431	0.000061399586	11/12	12/12
$N21, a5$	0.000730337378	0.000577315629	0.000286392554	12/12	12/12
$N33, a2$	0.000175745638	0.000185139622	0.000062518321	11/12	12/12
$N35, a2$	0.000402743572	0.000404525079	0.000054268499	12/12	12/12

Table 5: Primary gates in the r1 archive.

Gate	Status
Frozen instance family used	pass
All multiplicative orders exact	pass
All success-bin sets nonempty	pass
Response rank five and tangent dimension four	pass
Three-way budget exact	pass
Intrinsic preserves declared response	pass
Intrinsic preserves ideal multiplier	pass
Random comparator is tangent	pass
Reference aggregate confidence interval positive	pass
Random aggregate confidence interval positive	pass
SPSA aggregate confidence interval positive	pass
Intrinsic beats random on majority of instances	pass
Intrinsic beats SPSA on majority of instances	pass

5 Interpretation And Boundaries

The result is a reproducible empirical separation inside a declared synthetic model. It shows that exact-response fibres can contain task-relevant directions even when the ideal response is held fixed. It does not show that the same directions exist in a native compiled circuit, that a hardware implementation would benefit, that fault-tolerant resources are reduced, that cryptographic-scale factoring becomes easier, or that Shor order finding has an improved asymptotic complexity. The appropriate interpretation is navigability of a non-native exact-response fibre, not practical computational advantage.

6 Exploratory Geometry

The archived m3 PDF contained a quantitative Section VI, but the editable source and complete generation chain for those diagnostics are unavailable. This reconstructed m4 source therefore removes the unverified Hessian, 256-sample, infidelity-proxy, and preconditioning-ablation numerical claims. Future work should redeposit executable scripts, inputs, outputs, and hashes before using such geometric diagnostics as quantitative evidence.

A Primary Gates

The r1 archive reports the following primary checks as true: frozen instance family used; all multiplicative orders exact; all success-bin sets nonempty; response rank five and tangent dimension four; three-way budget exact; intrinsic declared response preserved; intrinsic ideal multiplier preserved; random comparator tangent; aggregate intrinsic-vs-reference confidence interval positive; aggregate intrinsic-vs-random confidence interval positive; aggregate intrinsic-vs-SPSA confidence interval positive; intrinsic beats random on a majority of instances; and intrinsic beats SPSA on a majority of instances.

B Reproducibility Correction

The v0.9.4-r1 archive retains the original v0.9.4 driver, protocol, result, stdout, tag, and release unchanged, and adds a corrective driver, explicit tangent basis, environment/protocol files, and a full 72-record result JSON. The correction does not attempt to recreate the undeclared v0.9.4 SPSA coordinate basis. It instead defines the SPSA coordinates explicitly and reruns the full 72-record benchmark. The r1 SPSA values are therefore not a bitwise reproduction of the old SPSA archive values.

C Code And Data Availability

Code, the frozen v0.9.4 protocol, complete per-meta records, reference stdout, integrity hashes, and r1 corrective artifacts are available in the `geometric-flow-qc` repository. The frozen computational tag is `shor-multinstance-fibre-v0.9.4`. The archived m3 PDF has SHA-256 `a258dc5f0ad8b7103794d7af5b8c98d8c56f5e7131eccf1dddc6aad51273913f`. The r1 result JSON has SHA-256 `3470ede84497391c8a99241db57aead6b0a642ee77a97ff27fed927e41f8aabf`.

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